

# Khaled Lakhdher

Computer Engineer | AI & Data Science | Full-Stack AI Developer

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🌐 [khaled-lakhdher-portfolio.vercel.app](https://khaled-lakhdher-portfolio.vercel.app)

## PROFILE

AI and data engineer who turns models and data into dependable products. I build LLM and RAG systems, multi-agent platforms, and end-to-end ML pipelines, with a metrics-first focus on data quality, evaluation, and clean deployment. Comfortable across the stack: Python, SQL, FastAPI, and Docker.

## EXPERIENCE

### AI Engineer - Final Year Project (Graduation)

Feb 2026 - June 2026 · Tunisia

CTT - Carthage Training & Technologies

- Designed and developed **Hotel Mind**, a conversational hotel-booking assistant on a grounded RAG architecture connected to a real-time SOAP API, with a hallucination-mitigation layer that verifies every factual claim.
- Delivered a containerized full-stack solution reaching nDCG@5 0.82, MRR@5 0.90, and a 100% success rate on a benchmark query set.

### Artificial Intelligence Developer Intern

July 2025 - September 2025 · Hammamet

CTT - Carthage Training & Technologies

- Fine-tuned a Hugging Face NLP model to classify customer reviews into three sentiment classes, improving accuracy by 25% over the baseline.
- Built a Django backend to manage customer data and automate export of analysis results to Excel for reporting.
- Owned the end-to-end Python pipeline for preprocessing, training, and integration.

## PROJECTS

### Nexus AI - Artificial Intelligence Employee Management Platform

- Engineered a multi-tenant platform (FastAPI, PostgreSQL/pgvector, Redis, Next.js) governing LLM agents with IAM permissions, vector memory, multi-agent orchestration over a Redis message bus, and real-time WebSocket monitoring, behind a provider-agnostic model gateway (Anthropic, OpenRouter, Gemini).

### Personalized Voice Assistant

- Fine-tuned Whisper (STT) and SpeechT5 (TTS) with CSM-1B voice cloning to build a personalized voice assistant, cutting response time by 30%.

### CodeReview Evaluation - Benchmarked LLM Code Reviewer

- Developed an LLM code-review and evaluation framework on the BugsInPy benchmark (500 real bugs); using an LLM-as-judge, showed a single detection score overstates capability by  $\sim 1.8\times$  (78% localization vs. 43% diagnosis) across 3 model tiers, with a live Streamlit demo.

## CERTIFICATIONS

- Fundamentals of Deep Learning - NVIDIA (Nov 22, 2022)
- Building Transformer-Based Natural Language Processing Applications - NVIDIA (Dec 19, 2025)
- Applications of AI for Predictive Maintenance - NVIDIA (Oct 27, 2025)

## SKILLS

**Programming Languages:** Python · Java · C · SQL · JavaScript · TypeScript

**Frontend Technologies:** React.js · React Native · HTML/CSS · Tailwind CSS

**AI Libraries & Frameworks:** Transformers · PyTorch · TensorFlow · Keras · Hugging Face · scikit-learn · Pandas · NumPy · Matplotlib

**Development Frameworks & Tools:** Gradio · Streamlit · Flask · Django · Zeep · LangSmith · Docker · Postman · Git · GitHub

**Concepts & Technologies:** Artificial Intelligence (AI) · Machine Learning (ML) · Deep Learning (DL) · Large Language Models (LLMs) · RAG (Retrieval-Augmented Generation) · AI Agents · Agentic AI · Tool Calling · Prompt Engineering · LLM Evaluation · LLM-as-a-Judge · Natural Language Processing (NLP) · Automatic Speech Recognition (ASR) · GANs · Semantic Search · Embeddings · NoSQL Databases · Cloud Computing

## EDUCATION

### Computer Engineering - AI & Data Science Specialization

2023 - 2026

EPI - International Polytechnic School · Sousse, Tunisia

### Preparatory Cycle

2021 - 2023

EPI - Private International Polytechnic School · Sousse, Tunisia

## LANGUAGES

**French:** DELF B2

**English:** TOEIC (Listening & Reading) - 825/990